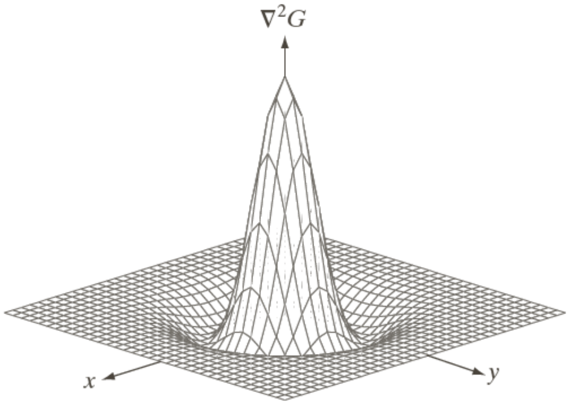
10.2.6 More Advanced Techniques for Edge Detection

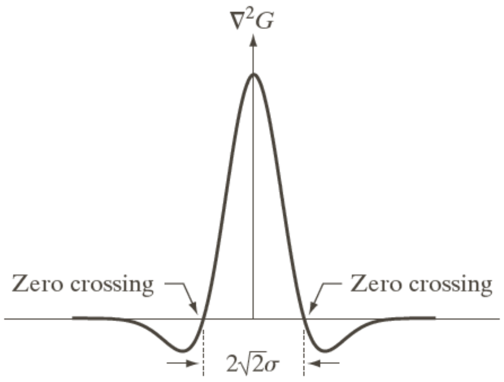
- The Marr-Hildreth edge detector

$$G(x,y)=e^{-\frac{x^2+y^2}{2\sigma^2}}$$

$$\nabla^2 G(x,y)=\left[\frac{x^2+y^2-2\sigma^2}{\sigma^4}\right]e^{-\frac{x^2+y^2}{2\sigma^2}}$$

The Marr-Hildreth edge detector





0	0	−1	0	0
0	−1	−2	−1	0
−1	−2	16	−2	−1
0	−1	−2	−1	0
0	0	−1	0	0

a	b
c	d

FIGURE 10.21 (a) Three-dimensional plot of the *negative* of the LoG. (b) Negative of the LoG displayed as an image. (c) Cross section of (a) showing zero crossings. (d) 5×5 mask approximation to the shape in (a). The negative of this mask would be used in practice.

$$\nabla^2 G(x,y) = \left[\frac{x^2 + y^2 - 2\sigma^2}{\sigma^4}\right] e^{-\frac{x^2+y^2}{2\sigma^2}}$$

The Marr-Hildreth edge detector

$$g(x,y) = [\nabla^2 G(x,y)] * f(x,y)$$

$$g(x,y) = \nabla^2 [G(x,y) * f(x,y)]$$

The Marr-Hildreth edge detector

- Step 1: filter the input image with Gaussian lowpass filter;
- Step 2:Compute the Laplacian of the image resulting from step 1;
- Step 3: find the zero crossing of the image from step 2;

Canny edge detector

- Step 1:Smooth the input image with a Gaussian filter;
- Step 2:Compute the gradient magnitude and angle image;
- Step 3:Apply non-maxima suppression to the gradient magnitude image;
- Step 4: use double thresholding and connectivity analysis to detect and link edge;

10.3 Thresholding

- 10.3.1 Foundation
  - The basic of intensity thresholding

$$g(x,y)=\begin{cases}1, & f(x,y)>Thr\\ 0, & f(x,y)<=Thr\end{cases}$$

10.3 Thresholding

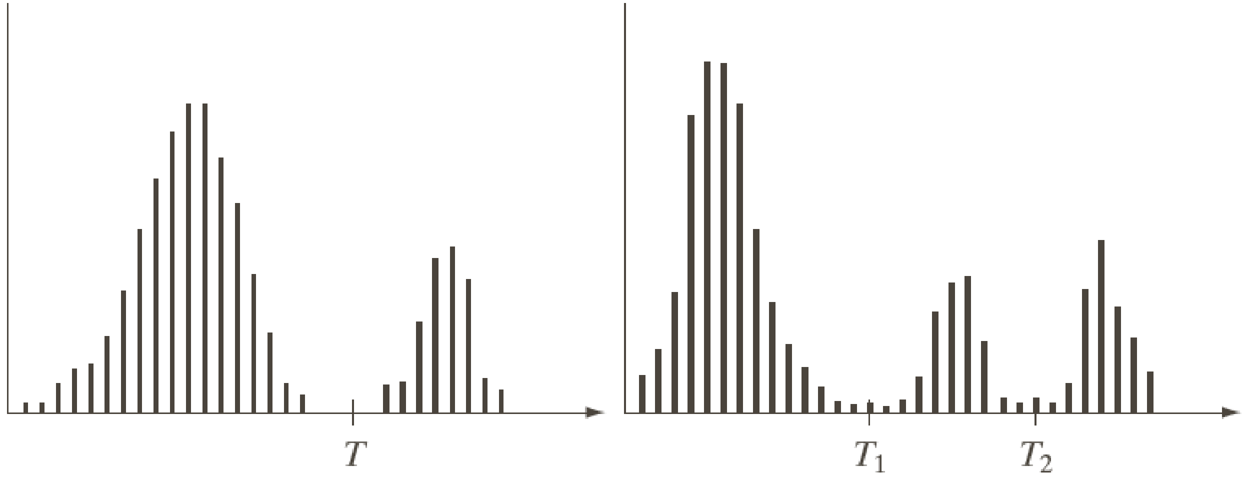
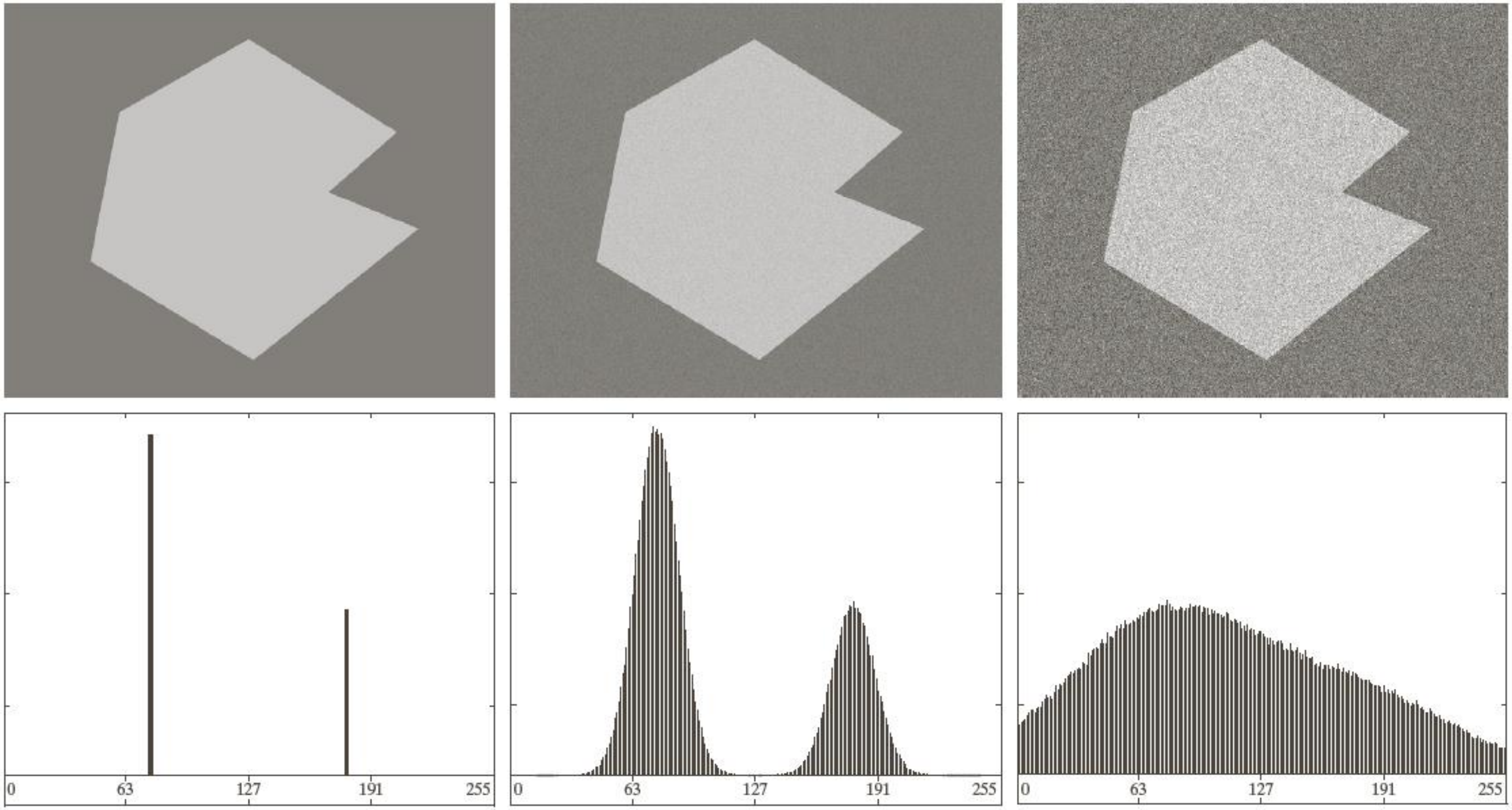
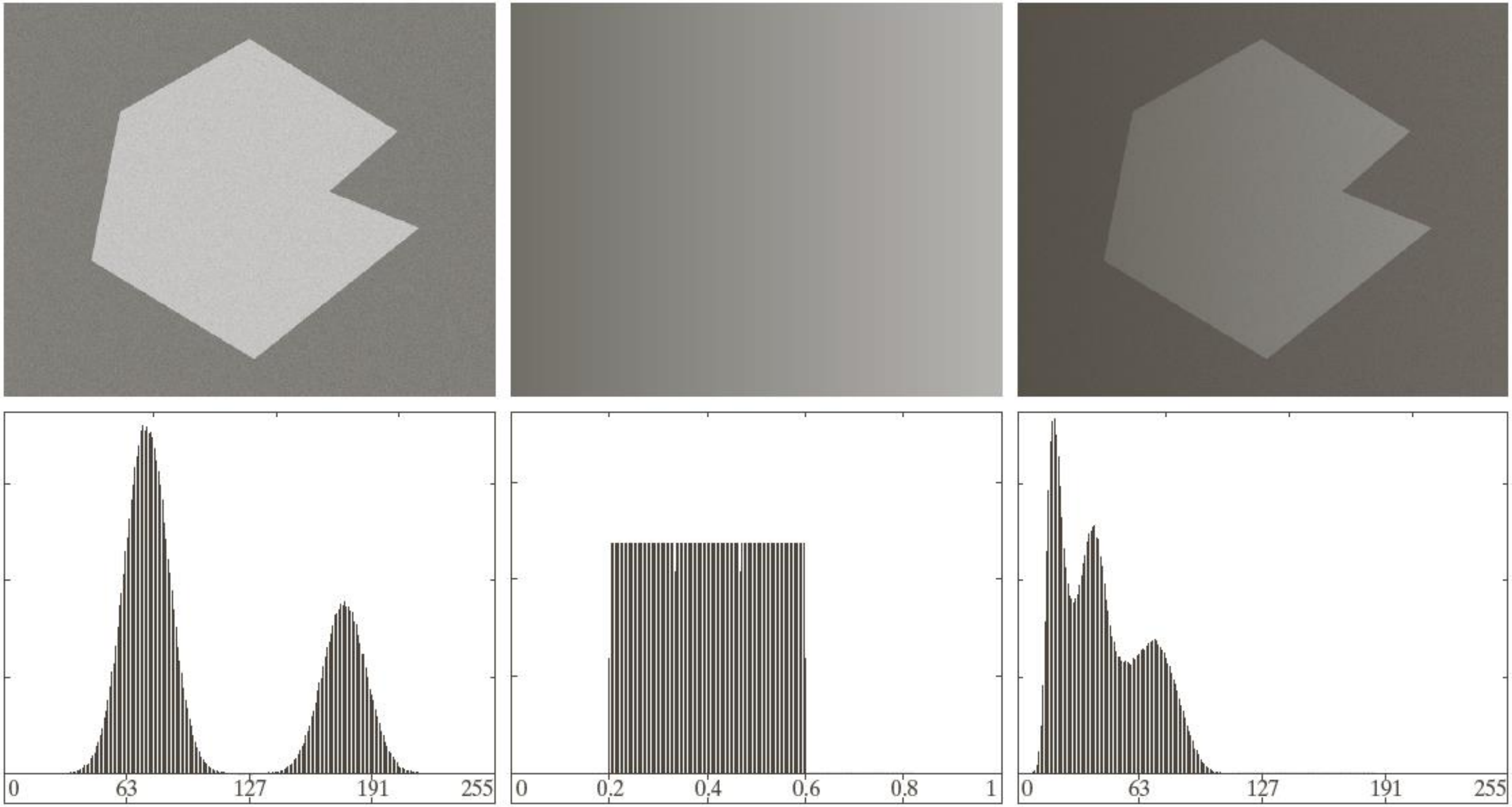


FIGURE 10.35 Intensity histograms that can be partitioned (a) by a single threshold, and (b) by dual thresholds.

10.3 Thresholding



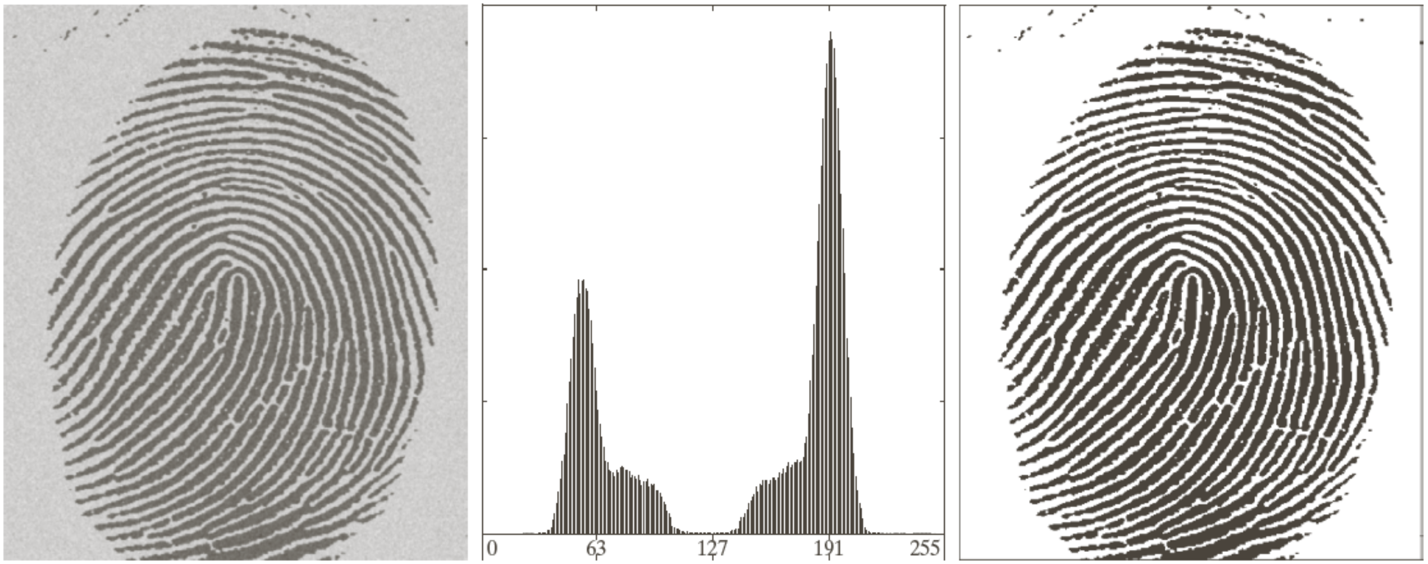
10.3 Thresholding



10.3.2 Basic Global Thresholding

- Step 1:select an initial estimate for the global threshold, T
- Step 2:segment the image using T in 2 Groups:$G1=\{(x,y)|f(x,y)>Thr\},G2=\{(x,y)|f(x,y)<=thr)\}$
- Step 3:compute the average intensity values m1 for G1 and m2 for G2
- Step 4: compute a new threshold value: $T=(m1+m2)/2$;
- Repeat step 2 to step 4,until $Abs(T\_new - T\_old)<\Delta T$

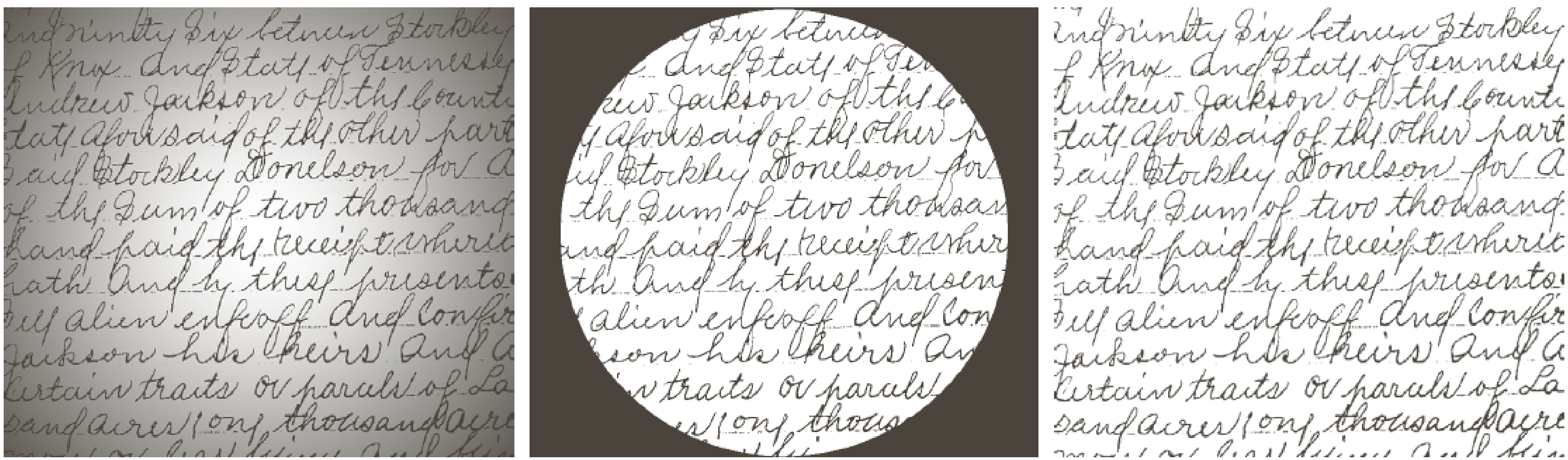
10.3.2 Basic Global Thresholding



a b c

FIGURE 10.38 (a) Noisy fingerprint. (b) Histogram. (c) Segmented result using a global threshold (the border was added for clarity). (Original courtesy of the National Institute of Standards and Technology.)

Local Threshoding



a b c

FIGURE 10.49 (a) Text image corrupted by spot shading. (b) Result of global thresholding using Otsu's method. (c) Result of local thresholding using moving averages.

Connected-component labeling

Input

0	0	0	0	0	0	0	0
0	0	ff	0	0	ff	ff	0
0	ff	ff		0	ff	ff	0
0	ff	ff	0	0	ff	ff	0
0	0	0	0	0	ff	ff	0
0	ff	ff	ff	0	ff	ff	0
0	ff	ff	0	0	0	ff	0
0	0	0	0	0	0	0	0

Output

0	0	0	0	0	0	0	0
0	0	1	0	0	2	2	0
0	1	1		0	2	2	0
0	1	1	0	0	2	2	0
0	0	0	0	0	2	2	0
0	3	3	3	0	2	2	0
0	3	3	0	0	0	2	0
0	0	0	0	0	0	0	0

Connected-component labeling

- Recursive function

```
void Label(byte [,] f, int x, int y, byte L)
{
        f[x,y] = L;

        if (f[x+1,y]==255) Label(f,x+1,y,L);
        if (f[x-1,y]==255) Label(f,x-1,y,L);
        if (f[x,y+1]==255) Label(f,x,y+1,L);
        if (f[x,y-1]==255) Label(f,x,y-1,L);
}
```

Connected-component labeling

- Application
  - Number of Connected-Components = Lmax-Lmin+1
  - Separate each Connected-component